

Continuous Charge Distributions

1 Overview

Previously, you found the electric field at a location in space due to more than one point charge by finding the electric field due to each charge and then vectorially summing the fields to get the total field (this is “using superposition”).

This process requires a significant amount of calculation if there are many charges. To reduce the number of calculations when charges are closely spaced, we sometimes assume they are continuously distributed; in this case, to compute the electric field, we need to evaluate an integral rather than a sum with many terms.

Section 21.5 of the textbook gives three examples of finding $\vec{E}$ for charges that are continuously and uniformly distributed: 1. along a straight line, 2. on a circle, and 3. on a disk. If you read the textbook examples and the lecture notes, you should be able to identify the following steps (not necessarily in this order).

1. Identify answer features (e.g., what should the direction of your final answer for $\vec{E}$ be?).
2. Find $d\vec{E}$ (or dE and its direction) for a dQ on the charged object. Use a diagram to check the direction.
3. Find dQ in terms of coordinates (e.g., dx , dy , r , $d\theta$, etc.).
4. Simplify $d\vec{E}$ (if possible) using symmetry arguments.
5. Integrate $d\vec{E}$.
6. Check answer features.

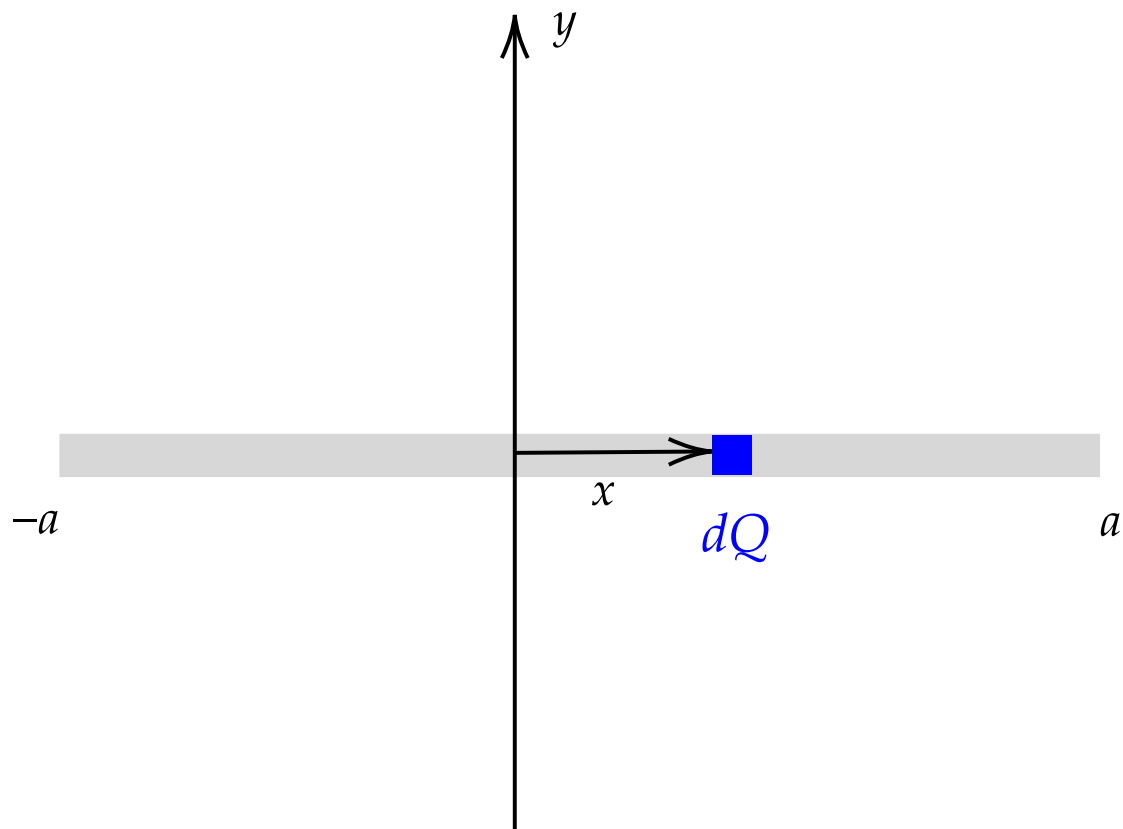
In this activity, you will explicitly address all of these steps for charges that are uniformly and continuously distributed along a straight line.

2 Finite Line of Charge

The following diagram shows a differential charge dQ located on the x -axis. Recall that the general equation for the electric field due to a point charge (with q replaced with dQ and $\vec{E}$ replaced with $d\vec{E}$) is

$$d\vec{E} = k \frac{dQ}{r^2} \hat{r}, \text{ which has magnitude } dE = k \frac{dQ}{r^2}$$

(In a previous activity, you found the components of $\vec{E}$ using two methods that are used in the textbook. You may use either method for this problem.)



1. Draw the expected directions of dE_x and dE_y at a location on the $+y$ -axis on the diagram given the location of dQ shown on the figure (assume dQ is positive).
2. In the diagram, the differential charge dQ is at a positive x . At the same location on the $+y$ -axis, draw the expected direction, using dotted lines for dE_x and dE_y , if dQ is at $-x$.
3. Based on your diagrams, do you expect any component of the electric field due to dQ at $+x$ will cancel that due to dQ at $-x$?
4. Find equations for the electric field components dE_x and dE_y at any location on the y -axis in terms of dQ , x , y , and k .

5. Suppose the charge dQ is at $x = a$ and you want to know the electric field at $y = a$. Plug these values into your equations for dE_x and dE_y . Are the signs of dE_x and dE_y consistent with what you drew for part 1.? Repeat supposing the charge dQ is at $x = -a$ and you want to know the electric field at $y = a$. Are the signs of dE_x and dE_y for this dQ consistent with what you drew for part 2.?
6. If the differential charge dQ is at $x = 0$, do you expect dE_x or dE_y to be zero? Is your answer consistent with the result of plugging in $x = 0$ into your equations for dE_x and dE_y ?

If the charge per unit length on the line is λ , then we can write $dQ = \lambda dx$. To find E_x and E_y , integrate dE_x and dE_y from $x = -a$ to a .

7. Write down the integrals that must be evaluated to find E_x and E_y . Do you expect either of the integrals to be zero?

8. From an integral table (or using trig substitutions), we know $\int \frac{du}{(u^2 + a^2)^{3/2}} = \frac{u}{a^2 \sqrt{u^2 + a^2}} + C$.
Use this to find E_y .

9. If y is negative (for example, $y = -a$), is your answer for E_y positive or negative? Is this consistent with what you expect based on the given diagram?
10. The total charge on the line is $Q = 2a\lambda$ (length $\times$ charge/length). Suppose $y \gg a$ so you can replace $a^2 + y^2$ with y^2 . Is your equation for E_y consistent with the electric field for a point charge $Q = 2a\lambda$ at the origin?

3 Long Line of Charge

Using the result of the previous problem, we can find E_y when $a \gg y$, which corresponds to a long line of charge. The result is

$$E_y = \frac{2\lambda k}{y}$$

This equation can be generalized to apply to any long, straight, and uniformly charged line of charge:

$$E = \frac{2\lambda k}{r}$$

where r is the perpendicular distance from the line. The direction of $\vec{E}$ is perpendicular to the line with a direction that depends on the sign of the charge density λ : if λ is positive, $\vec{E}$ points away from the line; if λ is negative, $\vec{E}$ points towards the line.

A long line of charge lies along the line $y = b$.

1. Find the electric field magnitude and direction at the origin.
2. Find the electric field magnitude and direction at $(x, y) = (b, 0)$.